

UniSeek

Falsifiable Hypothesis and AI Agent

Purpose: Define the core product assumption UniSeek must prove and the role of the AI system designed to prove it.

1. Falsifiable Hypothesis

A falsifiable hypothesis is a specific belief about the product that can be tested and proven wrong. For UniSeek, the central question is not whether AI can recommend colleges. General-purpose chatbots already do that. The real question is whether UniSeek creates enough additional value for students to prefer it.

Students will find UniSeek's guided, sourced college recommendations more useful and trustworthy than recommendations from a general-purpose chatbot.

How to test it

- Give students a recommendation list produced by UniSeek.
- Give the same students a recommendation list produced by ChatGPT or Claude.
- Hide which tool produced each result.
- Ask which list they find more useful, trustworthy, and actionable.

Outcome	Threshold	Meaning
Success	UniSeek wins at least 60%	The product is meaningfully outperforming the free alternative.
Kill signal	UniSeek wins 50% or less	The current experience is not creating enough added value.

What failure would mean

A failed test would not automatically mean the entire idea is dead. It would mean that at least one core assumption needs revision. Possible problems could include the intake questions, college data quality, recommendation logic, explanation format, or target audience.

Why this is the right hypothesis

UniSeek must prove that guided intake, stronger college data, citations, and structured explanations create a result that students value more than a free-form chatbot response. That is the product's riskiest

and most important assumption.

2. UniSeek's AI Agent

The AI agent is the part of UniSeek that turns a student's profile and UniSeek's college data into structured, transparent recommendations. It should not behave like an open-ended chatbot. It should operate as a constrained college-research and matching system.

What the agent receives

- Academic information: GPA, course rigor, test scores, and intended majors.
- College preferences: location, campus size, school type, cost, and financial priorities.
- Qualitative information: extracurriculars, interests, values, social preferences, and career goals.
- Risk preference: how ambitious or conservative the student wants the list to be.
- UniSeek's college corpus: official admissions and outcomes data plus corroborated qualitative information about campus culture.

What the agent does

1. Interpret the student profile	Translate intake responses into meaningful matching factors. For example, a robotics background may increase the importance of project-based engineering opportunities, while a student who dislikes intense competition may prefer a more collaborative academic culture.
2. Determine factor importance	Separate must-haves, strong preferences, flexible preferences, and acceptable tradeoffs. Affordability might be a hard requirement, while warm weather may only be a preference.
3. Retrieve eligible colleges	Search UniSeek's college corpus instead of relying only on model memory. Apply hard constraints first, then rank the remaining colleges using softer preferences.
4. Evaluate fit	Assess academic, financial, environmental, social, and extracurricular fit. Use explainable labels such as High Match, Medium Match, and Low Match instead of one unexplained numerical score.
5. Evaluate attainability	Classify schools separately as Reach, Target/Match, or Likely/Safety. Fit and admissions likelihood are different, and neither should be presented as a guarantee.
6. Generate structured output	Return approved response types such as why the school was recommended, why it may not fit, admissions outlook, major tradeoffs, confidence level, and visible sources.

7. Show uncertainty

Clearly separate official facts, qualitative consensus, model interpretation, and missing information. Low-data schools should receive a low-confidence label rather than a confident guess.

What the AI agent is not

- A replacement for an admissions officer.
- A guarantee of acceptance.
- A college-ranking oracle.
- A generic chat assistant.
- A system that makes the final college decision for the student.

The agent's role is to research, evaluate, explain, and narrow options - not make the final decision for the student.

3. How They Connect

The falsifiable hypothesis defines what UniSeek must prove. The AI agent defines how UniSeek attempts to prove it.

Falsifiable hypothesis	Students will prefer a guided, transparent, evidence-backed recommendation experience over a general chatbot.
AI agent	Creates that experience through guided profile interpretation, college retrieval, structured matching, admissions categorization, cited evidence, tradeoffs, and honest uncertainty.

One-sentence definition:

UniSeek's falsifiable hypothesis is that students will trust and prefer structured, source-backed college recommendations over free-form chatbot recommendations; its AI agent tests that hypothesis by converting a guided student profile into evidence-based college matches, admissions categories, tradeoffs, confidence labels, and cited explanations.